Foundation for Advancement of Science & Technology

HW # 25

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(a)

$$A = \begin{bmatrix} -4 & 2 \\ 3 & 1 \end{bmatrix} \qquad x_0 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

i) $|A - \lambda I| = 0$

$$\begin{vmatrix} -4-\lambda & 2 \\ 3 & 1-\lambda \end{vmatrix} = 0$$

$$(-4-\lambda)(1-\lambda) - 6 = 0$$

$$-4 + 3\lambda + \lambda^2 - 6 = 0$$

$$\lambda^2 + 3\lambda - 10 = 0$$

$$\lambda = \frac{-3 \pm 7}{2}$$

$\lambda = 2$ or $\lambda = -5$

$$E_2 = \text{span}\left\{\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right\}$$

$$E_{-5} = \text{span}\left\{\begin{bmatrix} -2 \\ 1 \end{bmatrix}\right\}$$

for $\lambda = 2$

$$\left[\begin{array}{cc|c} -6 & 2 & 0 \\ 3 & -1 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} -3 & 1 & 0 \\ 3 & -1 & 0 \end{array}\right]$$

$$\sim \left[\begin{array}{cc|c} -3 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$-3x_1 + x_2 = 0$$

$$x_1 = 1/3\, x_2$$

$$x_2 = x_2$$

$$x = \begin{bmatrix} 1/3 \\ 1 \end{bmatrix} x_2 \Rightarrow x = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

for $\lambda = -5$

$$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ 3 & 6 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$x_1 + 2x_2 = 0$$

$$x_2 = x_2$$

$$\begin{bmatrix} -2 \\ 1 \end{bmatrix} x_2$$

ii)

$$x_0 = c_1 v_1 + c_2 v_2 \longrightarrow (i)$$

$$x_0 = v_1 + v_2$$

$$Ax_0 = Av_1 + Av_2$$

$$Ax_0 = \lambda_1 v_1 + \lambda_2 v_2 \qquad \because Av_1 = \lambda_1 v_1$$

$$A^2 x_0 = A\lambda_1 v_1 + A\lambda_2 v_2$$

$$A^2 x_0 = \lambda_1^2 v_1 + \lambda_2^2 v_2$$

$$A^n x_0 = c_1 \lambda_1^n v_1 + c_2 \lambda_1^n v_2 \rightarrow (ii)$$

By (i)

$$\begin{bmatrix} -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & -2 & -1 \\ 3 & 1 & 4 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -2 & -1 \\ 0 & 7 & 7 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -2 & -1 \\ 0 & 1 & 1 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array}\right]$$

$$c_1 = 1$$

$$c_2 = 1$$

By (ii)

$$A^{10} x_0 = (1)(-5)^{10} \begin{bmatrix} -2 \\ 1 \end{bmatrix} + (1)(2)^{10} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$A^{10} x_0 = \begin{bmatrix} -2 \times 5^{10} \\ 5^{10} \end{bmatrix} + \begin{bmatrix} 2^{10} \\ 3 \times 2^{10} \end{bmatrix}$$

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(iii) $A^k x_0 = c_1 \lambda_1^k v_1 + c_2 \lambda_2^k v_2$

if $k \to \infty$

the values of λ will become extremely large as the λ's are in power of k.

(b)

$x(t) = c_1 v_1 e^{\lambda_1 t} + c_2 v_2 e^{\lambda_2 t}$

$x(t) = \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{2t} + \begin{bmatrix} -2 \\ 1 \end{bmatrix} e^{-5t}$

(c)

$x_n - 4x_{n-1} + 3x_{n-2} = 0$

$1\lambda^2 - 4\lambda + 3 = 0$

$\lambda = \dfrac{4 \pm \sqrt{16-12}}{2} = \dfrac{4 \pm 2}{2} = 3$ or 1

$x_0 = c_1 \lambda_1^n + c_2 \lambda_2^n \Rightarrow 0 = c_1 3^0 + c_2 \Rightarrow c_1 + c_2 = 0$

$x_1 = c_1 \lambda_1^n + c_2 \lambda_2^n \Rightarrow 1 = c_1 3^1 + c_2 \Rightarrow 3c_1 + c_2 = 1$

$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 3 & 1 & 1 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & -2 & 1 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 0 & 1/2 \\ 0 & 1 & -1/2 \end{array}\right]$

$c_1 = 1/2$

$c_2 = -1/2$

$\Rightarrow c = \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix}$